# cs5800-hw$_{1e}xercises$

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## 1 Exercise 0.1

(a) $f = \Theta(g)$, omit constants, $n^1 = n^1$
(b) $f = O(g)$, given $n^a, n^b$, if $a < b$, then $O(n^a) < O(n^b)$
(c) $f = O(g)$, omit $n^1$, $logn < (logn)^2$
(d) $f = \Theta(g)$, omit multiplicative constants, both sides are $O(nlogn)$
(e) $f = \Theta(g)$, omit multiplicative constants, both sides are $O(logn)$
(f) $f = \Theta(g)$, make $n = 10^n$, two sides are $10n$ and $2n$, equals to each other.
(g) $f = \Omega(g)$, cancel out $n$, then $n^{0.01} > log^2 n$, polynomial dominates logarithm
(h) $f = \Omega(g)$, both sides multiply by $logn$, cancel out n, then $n > (logn)^3$, polynomial dominates logarithm
(i) $f = \Omega(g)$, polynomial dominates logarithm
(j) $f = O(g)$, both sides multiply by $logn$, then $(logn)^{logn+1} < n$, polynomial dominates logarithm
(k) $f = \Omega(g)$, polynomial dominates logarithm
(l) $f = O(g)$, make $n = 2^n$, then $2^{n/2} < 5^n$
(m) $f = O(g)$, omit $n$, base number make huge difference in exponential
(n) $f = \Theta(g)$, $2^{n+1} = 2^n \cdot 2$, cancel out $2^n$
(o) $f = \Omega(g)$, factorial dominates exponential
(p) $f = O(g)$, make $n = logn$, then $n^n/2^{n^2} = nlogn/n^2$, $nlogn < n^2$
(q) $f = \Theta(g)$, $\sum_{i=1}^{n} i^k = (1 + n^k) \cdot n/(k+1)$, omit lower level polynomial and multiplicative constants, $n^{k+1}/(k+1) - n/(k+1) = n^{k+1}$

## 2 Exercise 0.2

(1)
if $0 < c < 1$, then $c^{n-1} > c^{n-1} \cdot c$, means $c^{n-1} > c^n$
therefore, $1 > c^1 > c^2 > \cdots > c^n$
therefore, 1 is dominant
therefore, (a) is correct.

(2)
if c = 1, then $c^{n-1} = c^{n-1}$ * c $= c^n = 1$

therefore, $1 + c^1 + c^2 + ... + c^n = \sum_{i=1}^{n} 1 = n$
therefore, (b) is correct.

(3)
if $c > 1$, then $c^{n-1} < c^{n-1} \cdot c$, means $c^{n-1} < c^n$
therefore, $1 < c^1 < c^2 < \cdots < c^n$
therefore, $c^n$ is dominant.
therefore, (c) is correct.

# 3 Exercise 0.4

(a)
Assume $M = \begin{pmatrix} a & b \\ \\ c & d \end{pmatrix}$, $N = \begin{pmatrix} e & f \\ \\ g & h \end{pmatrix}$
$M \cdot N = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}$
total 4 additions and 8 multiplications.
For $X^n$, it takes $(4 \cdot 2^n - 4)$ additions and $2^{2n-1}$ multiplications

(b)
When $n\%2 = 0$, $X^n = X^{n//2} \cdot X^{n//2}$
When $n\%2 = 1$, $X^n = X^{n//2} \cdot X^{n//2} \cdot X$
Keep dividing until $n = 1$
total time complexity is O($logn$)